

Transformer-Based Customer Support Email Routing: A Comparative Study of GPT-2 Prompting, Parameter-Efficient Fine-Tuning, and DistilBERT

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Abstract

Efficient customer support depends on accurately routing incoming requests to the appropriate service department. As the volume of customer interactions continues to increase, manual ticket routing becomes labor-intensive, inconsistent, and difficult to scale. Recent advances in transformer-based language models have demonstrated strong capabilities in natural language understanding, making them suitable candidates for automating customer support workflows. This paper presents a comparative evaluation of three transformer-based approaches for intelligent customer support email routing: GPT-2 Prompting, GPT-2 with Low-Rank Adaptation (LoRA), and DistilBERT. A comprehensive preprocessing pipeline was developed using the Customer Support Tickets dataset from Hugging Face, including language filtering, department selection, text construction, label encoding, and stratified dataset splitting. The models were evaluated using accuracy, precision, recall, F1-score, confusion matrices, and inference time to provide a balanced assessment of predictive performance and computational efficiency. Experimental results show a consistent improvement across the three approaches. GPT-2 Prompting achieved an accuracy of 56.50%, GPT-2 with LoRA improved the accuracy to 65.86% while updating only 0.24% of the model parameters, and DistilBERT obtained the highest overall performance with an accuracy of 72.17%. These findings demonstrate that supervised transformer-based classification substantially outperforms prompt-based inference for customer support email routing while highlighting the trade-offs between parameter efficiency, computational cost, and predictive performance.

Keywords— Natural Language Processing, Large

Language Models, Transformer Models, GPT-2, DistilBERT, LoRA, Customer Support Automation, Email Routing, Text Classification, Hugging Face.

1 Introduction

Customer support services represent one of the most critical communication channels between organizations and their customers. Every day, companies receive large volumes of emails covering technical issues, billing inquiries, product information, sales requests, and general customer questions. Accurately routing these requests to the appropriate department is essential for reducing response times, improving service quality, and increasing customer satisfaction. However, manual ticket routing is often time-consuming, inconsistent, and prone to human error, particularly in organizations that process thousands of support requests daily.

Recent developments in large language models and transformer architectures have significantly advanced the field of Natural Language Processing. Models such as GPT-2, BERT, and DistilBERT have demonstrated strong capabilities in understanding contextual language representations and performing a wide range of language understanding tasks. These models provide an opportunity to automate customer support workflows by learning semantic relationships within customer messages and predicting the most appropriate support department.

Despite these advances, selecting an appropriate transformer-based approach requires balancing predictive performance, computational efficiency, and training cost. Prompt-based methods require little or no task-specific training but often struggle with complex multi-class classification tasks. Parameter-

efficient fine-tuning techniques, such as Low-Rank Adaptation, reduce computational requirements by updating only a small subset of model parameters while preserving much of the performance of full fine-tuning. Fully supervised transformer models, such as DistilBERT, generally achieve stronger predictive performance but require greater computational resources during training.

To investigate these trade-offs, this study performs a comparative evaluation of three transformer-based approaches for intelligent customer support email routing using a common preprocessing pipeline and evaluation framework. The experiments are conducted on the Customer Support Tickets dataset available through Hugging Face, with the task restricted to five representative support departments: Billing and Payments, Customer Service, General Inquiry, Sales and Pre-Sales, and Technical Support.

Unlike studies that evaluate only a single architecture, this work directly compares three complementary approaches representing zero-shot prompting, parameter-efficient fine-tuning, and fully supervised sequence classification under identical experimental conditions. This comparison provides practical insights into the advantages and limitations of each approach for real-world customer support automation.

1.1 Main Contributions

The primary contributions of this work are summarized as follows:

- A complete transformer-based pipeline for intelligent customer support email routing, including data preprocessing, model training, evaluation, and comparative analysis.
- A comprehensive experimental comparison of GPT-2 Prompting, GPT-2 with LoRA, and DistilBERT using the same preprocessed dataset and evaluation metrics.
- A reproducible preprocessing and evaluation framework implemented using the Hugging Face ecosystem and modern NLP libraries.
- An empirical analysis of the trade-offs between prompt-based inference, parameter-efficient fine-tuning, and fully supervised transformer-based

text classification for customer support automation.

The remainder of this paper is organized as follows. Section 2.5 reviews related work on transformer-based text classification and customer support automation. Section 3 describes the dataset and preprocessing pipeline. Sections 7–9 present the three transformer-based approaches. Section 10 reports the experimental results and comparative analysis. Finally, Sections 11–14 discuss the findings, limitations, future research directions, and conclusions.

2 Related Work

Recent advances in Natural Language Processing have been largely driven by transformer-based architectures, which have significantly improved performance across a wide range of language understanding and generation tasks. The development of large pre-trained language models has enabled researchers to leverage transfer learning for downstream applications such as text classification, question answering, sentiment analysis, machine translation, and conversational AI. This section reviews the key developments related to transformer architectures, large language models, parameter-efficient fine-tuning, and intelligent customer support automation, providing the context for the comparative study presented in this work.

2.1 Transformer Models for Natural Language Processing

The introduction of the Transformer architecture by Vaswani et al. [4] fundamentally changed the field of Natural Language Processing by replacing recurrent neural networks with a self-attention mechanism capable of modeling long-range dependencies more efficiently. Unlike recurrent architectures that process sequences sequentially, transformers process input tokens in parallel, enabling faster training and improved scalability. This architecture became the foundation for many modern language models and established a new state of the art across numerous NLP benchmarks.

Transformer models generally consist of encoder and decoder components. Encoder-based models focus on learning contextual representations of text and

are particularly effective for language understanding tasks such as text classification and named entity recognition. Decoder-based models, on the other hand, are designed for autoregressive language generation and are widely used for text completion and conversational applications. The widespread adoption of transformer architectures has established transfer learning as a standard paradigm in NLP, where large models are first pre-trained on extensive text corpora and subsequently adapted to downstream tasks through fine-tuning.

2.2 Large Language Models

Large Language Models extend transformer architectures by scaling model size, training data, and computational resources to learn general-purpose language representations. Among the earliest influential autoregressive language models, GPT-2 demonstrated that large-scale unsupervised pre-training enables coherent text generation and zero-shot capabilities across multiple language tasks [?]. However, because GPT-2 was primarily designed for text generation rather than classification, its performance on structured classification problems often depends heavily on prompt design and output interpretation.

In contrast, encoder-based models such as BERT were specifically developed for bidirectional language understanding [1]. By jointly considering both left and right context during pre-training, BERT achieved substantial improvements in language understanding benchmarks and became a standard architecture for supervised text classification. DistilBERT, a compressed version of BERT introduced by Sanh et al. [3], preserves much of BERT’s predictive capability while reducing model size, inference time, and computational requirements. This balance between efficiency and accuracy has made DistilBERT a popular choice for practical NLP applications where computational resources are limited.

The complementary characteristics of GPT-2 and DistilBERT make them suitable candidates for comparative evaluation. GPT-2 represents a generative transformer capable of prompt-based inference, whereas DistilBERT represents an encoder-based transformer specifically optimized for discriminative sequence classification tasks.

2.3 Parameter-Efficient Fine-Tuning

Although full fine-tuning of transformer models generally provides strong predictive performance, updating every model parameter can require substantial computational resources and memory. As transformer models continue to increase in size, parameter-efficient fine-tuning techniques have emerged as practical alternatives that reduce training cost while maintaining competitive performance.

One of the most influential approaches is Low-Rank Adaptation (LoRA), proposed by Hu et al. [2]. Instead of modifying all model parameters, LoRA introduces small trainable low-rank matrices into selected transformer layers while keeping the original model weights frozen. This approach dramatically reduces the number of trainable parameters, decreases GPU memory consumption, and accelerates fine-tuning without requiring a separate copy of the entire model.

Because LoRA updates only a small fraction of the original parameters, it has become widely adopted for adapting large language models to downstream tasks under limited computational resources. In this study, LoRA is applied to GPT-2 to investigate whether parameter-efficient adaptation can substantially improve classification performance while maintaining computational efficiency.

2.4 Intelligent Customer Support Automation

Automatic routing of customer support requests has become an increasingly important application of Natural Language Processing due to the growing volume of customer interactions handled by modern organizations. Traditional ticket routing systems commonly relied on manually designed rules, keyword matching, or conventional machine learning algorithms such as Support Vector Machines, Naive Bayes, and Random Forest classifiers. While these methods provided reasonable performance for limited domains, they often struggled to capture contextual relationships within complex customer messages.

Transformer-based language models have significantly improved customer support automation by learning semantic representations that extend beyond individual keywords. Their contextual understand-

ing enables more accurate classification of customer requests, even when different users describe similar problems using different vocabulary or writing styles. Recent studies and practical systems have demonstrated the effectiveness of transformer models for helpdesk automation, intent detection, customer service chatbots, and email classification, highlighting their potential to reduce manual workload while improving routing accuracy and customer response time.

2.5 Research Gap

Existing research demonstrates the effectiveness of transformer-based language models for text classification and customer support automation. However, many studies primarily focus on a single transformer architecture or a single adaptation strategy. Comparatively fewer works provide a systematic evaluation of prompt-based inference, parameter-efficient fine-tuning, and fully supervised transformer classification under a common preprocessing pipeline and evaluation framework for customer support email routing.

This study addresses this gap by implementing and experimentally comparing three representative transformer-based approaches: GPT-2 Prompting, GPT-2 with Low-Rank Adaptation, and DistilBERT. By evaluating these models using the same dataset, preprocessing pipeline, and evaluation metrics, this work provides practical insights into the trade-offs between predictive performance, computational efficiency, and training requirements for intelligent customer support automation.

3 Dataset Description

The experiments presented in this study were conducted using the *Customer Support Tickets* dataset available through the Hugging Face Datasets repository. This dataset was selected because it closely resembles real-world customer support scenarios, containing customer-generated support requests together with their corresponding routing departments. The diversity of ticket categories, writing styles, and metadata makes the dataset well suited for evaluating transformer-based approaches to automatic email routing.

The original dataset contains 61,765 customer

Characteristic	Value
Dataset Source	Hugging Face
Total Samples	61,765
Features	16
Languages	English, German
Support Queues	52
Target Queues Used	5
Primary Input Fields	Subject, Body

Table 1: Summary of the original Customer Support Tickets dataset.

support tickets described by 16 attributes, including the ticket subject, message body, assigned support queue, language, ticket priority, ticket type, customer tags, and other metadata. The tickets are distributed across 52 customer support departments and are written in two languages: English and German. Since this work focuses on English-language customer support automation, only English tickets were considered during preprocessing.

Among the available attributes, the *subject* and *body* fields provide the primary textual information describing each customer request. These two fields contain complementary information: the subject typically summarizes the customer’s issue, while the message body provides additional context and detailed problem descriptions. Together, they form the input text used for all transformer-based classification models evaluated in this study.

Although the dataset contains a wide variety of customer support categories, this study focuses on five representative departments that frequently occur in practical customer service environments: Billing and Payments, Customer Service, General Inquiry, Sales and Pre-Sales, and Technical Support. Restricting the classification task to these five departments allows the study to investigate transformer-based email routing under a realistic multi-class classification setting while maintaining sufficient training samples for each category. This selection also enables a fair comparison of the three transformer-based approaches without introducing excessive class sparsity associated with less frequent departments.

The exploratory analysis revealed several characteristics relevant to the subsequent preprocessing stage. The dataset exhibits noticeable class imbalance across support queues, with Technical Support rep-

Dataset Split	Samples
Training	11,592
Validation	1,657
Testing	3,313
Total	16,562

Table 2: Final processed dataset split used for model training and evaluation.

representing the largest category. Furthermore, missing values were identified in the textual fields, including 5,299 missing subject entries and 2 missing message bodies. These observations motivated the preprocessing pipeline described in the following section.

Following language filtering, department selection, missing-value handling, and text construction, the dataset was reduced to 16,562 English customer support tickets. The processed dataset was subsequently divided into training, validation, and testing subsets using stratified sampling to preserve the class distribution during model training and evaluation.

The processed dataset serves as a common benchmark for all three transformer-based approaches evaluated in this work, ensuring that differences in predictive performance arise from the models themselves rather than inconsistencies in data preparation.

4 Exploratory Data Analysis

Exploratory Data Analysis (EDA) was performed to understand the characteristics of the customer support ticket dataset before preprocessing and model development. The primary objectives of this stage were to examine the distribution of support requests, evaluate data quality, identify missing values, analyze class imbalance, and assess the suitability of the dataset for transformer-based text classification.

The original dataset contains 61,765 customer support tickets represented by 16 attributes, including textual information and ticket metadata. The analysis confirmed that the dataset consists of two languages, German and English, distributed across 52 distinct customer support queues. Because the objective of this study is English-language customer support automation, only English tickets were considered during subsequent preprocessing.

Property	Value
Total Samples	61,765
Features	16
Languages	2
Support Queues	52
Missing Subject Values	5,299
Missing Body Values	2

Table 3: Dataset characteristics observed during exploratory analysis.

4.1 Dataset Overview

The initial inspection revealed that the dataset contains a diverse collection of customer support requests covering multiple service departments. In addition to the textual attributes, subject and body, the dataset includes metadata such as ticket priority, ticket type, language, tags, and department assignments.

4.2 Language Distribution

The language analysis showed that the dataset contains both English and German customer support requests. German tickets account for 33,504 records, while English tickets account for 28,261 records. Although both languages are well represented, this study focuses exclusively on English tickets to ensure linguistic consistency throughout model training and evaluation.

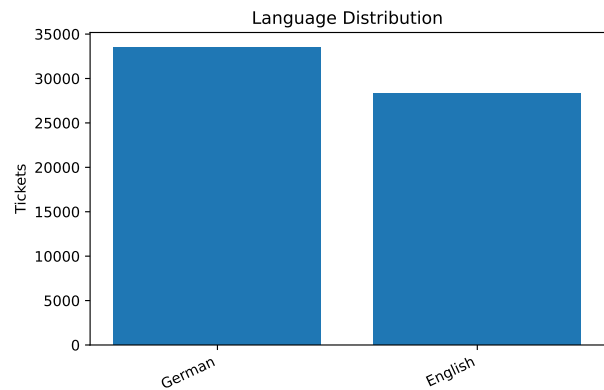


Figure 1: Language distribution of customer support tickets.

Restricting the dataset to a single language reduces linguistic variability and enables the models to

learn more consistent semantic representations.

4.3 Support Queue Distribution

The dataset contains customer requests distributed across 52 support departments, indicating a broad range of customer service scenarios. However, the class distribution is highly imbalanced, with several departments containing substantially more tickets than others. Technical Support is the most frequent category in the original dataset.

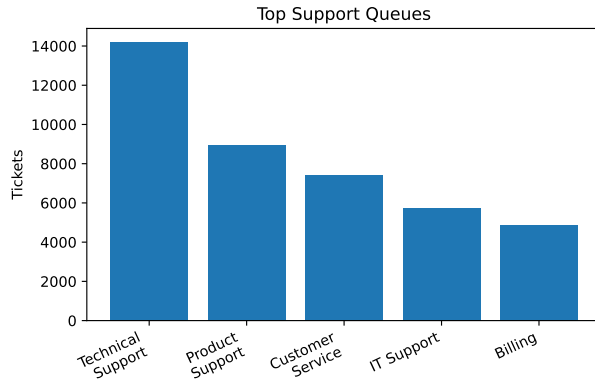


Figure 2: Distribution of the most frequent support queues in the original dataset.

The observed imbalance motivated the selection of five representative departments with sufficient training samples to support reliable model evaluation.

4.4 Missing Value Analysis

Data quality assessment identified missing values primarily within the textual fields. The analysis revealed 5,299 missing subject entries and 2 missing message bodies. Missing values were also observed in several metadata fields, but those attributes were not used as model inputs in this study.

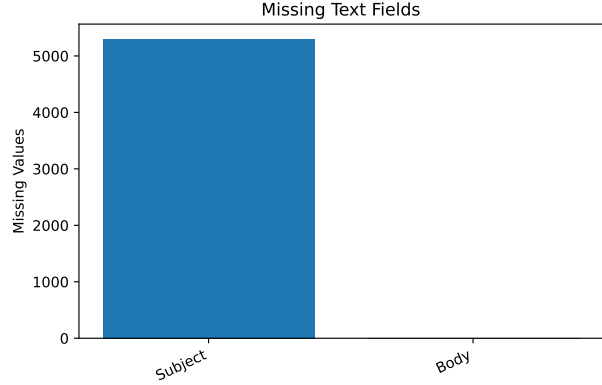


Figure 3: Missing values in the main textual fields.

Rather than removing records with missing textual fields, missing subjects and bodies were replaced with empty strings during preprocessing. This approach preserved valuable information contained in the remaining text while avoiding unnecessary loss of training data.

4.5 Target Department Distribution

Following language filtering, the study focused on five customer support departments: Billing and Payments, Customer Service, General Inquiry, Sales and Pre-Sales, and Technical Support. After filtering, the dataset contained 16,562 English customer support tickets.

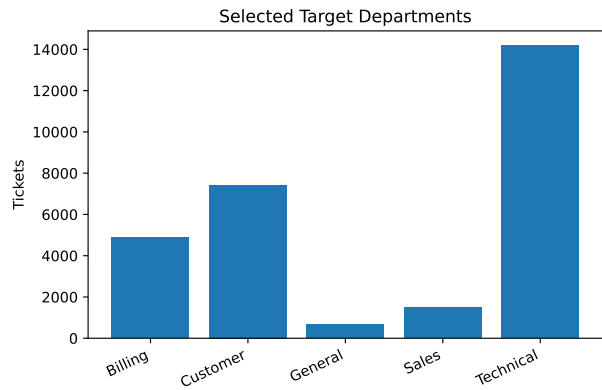


Figure 4: Distribution of selected target departments used in the classification task.

Although the selected departments remain imbal-

anced, each category contains sufficient samples to support supervised transformer training and evaluation.

4.6 Key Findings

The exploratory analysis provided several important insights that guided the design of the preprocessing pipeline and subsequent experiments. The dataset contains a large number of customer support requests suitable for transformer-based learning. Customer tickets are available in two languages, requiring language-specific filtering. The support queue distribution is highly imbalanced across the original 52 departments. Missing values are concentrated primarily within the subject field, while missing message bodies are extremely rare.

These observations directly informed the preprocessing strategy described in the following section, including language filtering, department selection, missing-value handling, and text construction.

5 Dataset Preprocessing

The customer support dataset required several preprocessing operations before it could be used for transformer-based text classification. The objective of this stage was to transform heterogeneous raw customer support tickets into a consistent representation suitable for supervised learning while preserving as much semantic information as possible. The preprocessing pipeline was designed to ensure that all three transformer models received identical input data, thereby enabling a fair comparison of their classification capabilities.

5.1 Language Filtering

The original dataset contains customer support tickets written in both English and German. Since multilingual training was outside the scope of this study, only English-language tickets were retained. Restricting the dataset to a single language reduces linguistic variability and prevents the models from learning language-specific patterns that are unrelated to the routing task. After language filtering, the dataset contained 28,261 English customer support tickets.

5.2 Department Selection

Although the original dataset contains 52 customer support queues, many categories contain relatively few training examples, resulting in severe class imbalance. To construct a practical multi-class classification task, the dataset was restricted to five representative departments: Billing and Payments, Customer Service, General Inquiry, Sales and Pre-Sales, and Technical Support. Following department selection, the dataset consisted of 16,562 customer support tickets.

5.3 Missing Value Handling

The exploratory analysis revealed missing values primarily within the textual attributes. Rather than removing these observations, missing textual fields were replaced with empty strings. This strategy preserves the available information contained in the remaining text while avoiding unnecessary reductions in training data.

5.4 Text Construction

Customer support requests typically consist of two complementary textual fields: a concise subject describing the issue and a more detailed message body providing additional context. To preserve information from both sources, these fields were concatenated into a single textual sequence using the following structure:

```
Subject: <ticket subject>  
Body: <ticket body>
```

5.5 Feature Selection and Label Encoding

Several metadata attributes included in the original dataset, such as ticket priority, tags, version information, ticket type, and predefined answers, were excluded from the classification experiments. Only the constructed text field and the corresponding department label were retained.

5.6 Dataset Splitting

To ensure reproducibility and unbiased model evaluation, the processed dataset was divided into training,

Label ID	Department
0	Billing and Payments
1	Customer Service
2	General Inquiry
3	Sales and Pre-Sales
4	Technical Support

Table 4: Department label encoding used for supervised classification.

validation, and testing subsets using stratified sampling. Stratification preserves the class distribution across all subsets, reducing the likelihood of biased performance estimates caused by uneven class representation.

5.7 Preprocessing Pipeline

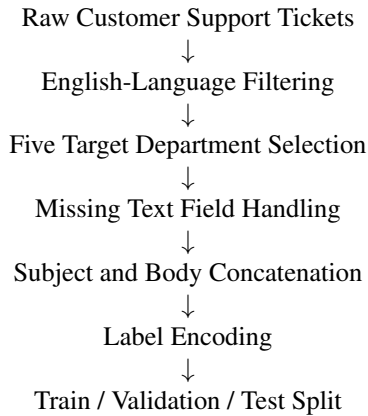


Figure 5: Dataset preprocessing pipeline used to transform raw customer support tickets into model-ready inputs.

The preprocessing pipeline transformed a heterogeneous customer support dataset into a standardized corpus suitable for transformer-based classification. Every model evaluated in this study received exactly the same processed dataset, ensuring that observed performance differences resulted from model architecture and learning strategy rather than inconsistencies in data preparation.

6 Methodology

This section describes the experimental framework used to evaluate transformer-based approaches for in-

telligent customer support email routing. To ensure a fair comparison, all models were trained and evaluated using the same preprocessed dataset, identical train-validation-test splits, and a common evaluation framework. The experiments were designed to compare three complementary paradigms for text classification: prompt-based inference, parameter-efficient fine-tuning, and supervised transformer learning.

6.1 Overall System Architecture

The proposed system follows a modular processing pipeline that transforms raw customer support requests into department predictions through preprocessing, model inference, evaluation, and comparative analysis.

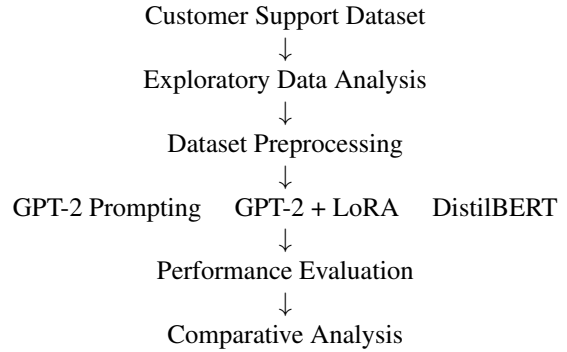


Figure 6: Overall experimental workflow for transformer-based customer support email routing.

All models were developed using the same processed dataset and evaluation utilities to ensure that performance differences reflected model architecture and adaptation strategy rather than differences in data preparation.

6.2 Experimental Environment

All experiments were implemented in Python using the Google Colaboratory environment with GPU acceleration. The implementation relies primarily on the Hugging Face ecosystem, which provides efficient tools for transformer-based model development, dataset management, and parameter-efficient fine-tuning.

Component	Specification
Programming Language	Python
Deep Learning Framework	PyTorch
Transformer Library	Hugging Face Transformers
Dataset Library	Hugging Face Datasets
Fine-Tuning Library	PEFT
GPU Environment	Google Colab Tesla T4
Evaluation Toolkit	Scikit-learn

Table 5: Experimental computing environment used in this study.

The experiments were conducted using a fixed random seed to improve reproducibility across pre-processing, dataset splitting, and model training.

6.3 Evaluation Metrics

The three transformer models were evaluated using identical quantitative metrics to enable objective comparison. Accuracy measures the proportion of correctly classified customer support tickets. Precision measures the proportion of predicted tickets that belong to the correct department, while recall measures the proportion of tickets belonging to a department that are correctly identified. The F1-score represents the harmonic mean of precision and recall.

For multi-class evaluation, weighted precision, recall, and F1-score were used to account for differences in class frequencies. In addition to these metrics, confusion matrices were used to analyze class-wise prediction performance. Computational efficiency was evaluated using inference time and the proportion of trainable parameters updated during model training.

6.4 Experimental Design

The objective of this study is not only to maximize classification accuracy but also to investigate the trade-offs between different transformer adaptation strategies. Accordingly, the experiments compare three representative approaches: GPT-2 Prompting, representing prompt-based inference without supervised training; GPT-2 with Low-Rank Adaptation, representing parameter-efficient fine-tuning; and DistilBERT, representing fully supervised transformer-based sequence classification.

All models were trained and evaluated using the same preprocessed dataset, identical class definitions, and consistent evaluation procedures. This experimental design isolates the influence of model architecture and training strategy while minimizing confounding factors introduced by differences in data preparation or evaluation methodology.

7 GPT-2 Prompting

7.1 Motivation

Prompt-based learning has emerged as an attractive alternative to conventional supervised learning because it enables pre-trained language models to perform downstream tasks with little or no task-specific parameter updates. Rather than training a dedicated classifier, prompt-based approaches reformulate classification as a text generation problem, allowing large language models to infer labels directly from natural language instructions.

Among autoregressive transformer models, GPT-2 represents one of the earliest and most influential generative language models. Although it was originally designed for open-ended text generation rather than classification, GPT-2 provides an effective baseline for investigating the capabilities and limitations of prompt-based inference for customer support email routing.

In this study, GPT-2 serves as a zero-shot baseline, allowing the subsequent LoRA and DistilBERT models to be evaluated relative to a model that receives no supervised task-specific training.

7.2 Prompt Design

Each customer support ticket was converted into a structured natural language prompt that instructs GPT-2 to predict the appropriate department. The prompt follows an instruction-style template containing the ticket subject, ticket body, and the list of possible departments.

You are an intelligent customer support routing assistant. Read the following customer request. Possible departments: Billing and Payments, Customer Service, General Inquiry, Sales and Pre-Sales, Technical Sup-

Component	Configuration
Model	GPT-2
Training	None
Adaptation	Zero-shot prompting
Input	Prompt containing subject and body
Output	Generated department label
Prediction	String matching with fallback
Framework	Hugging Face Transformers

Table 6: Configuration of the GPT-2 prompting baseline.

port. Respond using only the department name.

Unlike conventional classifiers, GPT-2 generates free-form text rather than fixed class labels. Consequently, the output must be interpreted and mapped to one of the predefined department categories.

7.3 Prediction Strategy

For each customer support ticket, GPT-2 generated a textual response conditioned on the constructed prompt. The generated output was compared against the predefined department names. If one of the valid department labels appeared in the generated response, that department was selected as the predicted class.

When GPT-2 failed to produce one of the expected department names, a fallback strategy assigned the prediction to Technical Support, the most frequently occurring department in the dataset. This strategy ensured that every customer support ticket received a valid prediction while preserving a fully automated inference pipeline.

7.4 Implementation

The GPT-2 baseline was implemented using the Hugging Face Transformers library. Because GPT-2 is used only for inference, no gradient updates or parameter optimization were performed.

7.5 Experimental Results

Table 7 summarizes the predictive performance of GPT-2 Prompting.

The experimental results indicate that GPT-2 Prompting provides a useful baseline but exhibits several

Metric	Value
Accuracy	56.50%
Precision	41.56%
Recall	56.50%
F1-score	45.65%

Table 7: GPT-2 Prompting performance on the evaluation subset.

limitations for multi-class customer support classification.

7.6 Discussion

Although GPT-2 possesses strong general language generation capabilities, it was not originally designed for discriminative classification tasks. Consequently, the model frequently generated responses that did not exactly correspond to one of the predefined department labels, requiring rule-based post-processing to convert generated text into valid class predictions.

Furthermore, GPT-2 demonstrated a noticeable tendency to over-predict Technical Support, reflecting both the imbalance of the evaluation dataset and the limitations of prompt-based label extraction. Similar confusion was observed between General Inquiry and Customer Service, suggesting that prompt-based inference alone provides limited discrimination between semantically related customer requests.

Despite these limitations, GPT-2 establishes an informative zero-shot baseline that highlights the potential benefits of task-specific adaptation. The results obtained in this experiment motivate the investigation of parameter-efficient fine-tuning through LoRA.

8 GPT-2 with Low-Rank Adaptation

8.1 Motivation

Although prompt-based inference enables pre-trained language models to perform downstream tasks without additional training, its effectiveness is often limited for structured multi-class classification problems. One straightforward solution is to fine-tune the entire transformer model. However, full fine-tuning requires updating millions of parameters, resulting in substantial computational cost, increased GPU memory consumption, and longer training times.

To address these limitations, recent research has introduced Parameter-Efficient Fine-Tuning methods that adapt large language models while updating only a small subset of their parameters. Among these approaches, Low-Rank Adaptation has become one of the most widely adopted techniques due to its simplicity, computational efficiency, and competitive predictive performance.

In this work, LoRA is applied to GPT-2 to investigate whether parameter-efficient adaptation can improve customer support email classification while maintaining a lightweight training process.

8.2 Low-Rank Adaptation

LoRA introduces trainable low-rank matrices into selected linear transformation layers of a pre-trained transformer while keeping the original model parameters frozen. Rather than modifying the complete weight matrices, LoRA learns small residual updates that approximate the required task-specific adaptations.

This strategy offers several practical advantages: substantially fewer trainable parameters, reduced GPU memory consumption, faster fine-tuning, preservation of the original pre-trained knowledge, and efficient adaptation to downstream NLP tasks. Because only a small number of additional parameters are optimized, LoRA provides an attractive balance between computational efficiency and predictive performance.

8.3 LoRA Configuration

The GPT-2 model was adapted using the Hugging Face PEFT library. LoRA adapters were inserted into the transformer attention layers while the original GPT-2 parameters remained frozen throughout training.

Only approximately 0.24% of the original GPT-2 parameters were updated during fine-tuning, illustrating the computational efficiency of the LoRA approach.

8.4 Training Procedure

The preprocessed dataset was tokenized using the GPT-2 tokenizer and divided into training, validation, and testing subsets using the stratified splits described

Parameter	Value
Base Model	GPT-2
Adaptation Method	LoRA
Task Type	Sequence Classification
LoRA Rank	8
LoRA Alpha	16
LoRA Dropout	0.10
Bias	None
Trainable Parameters	0.24%

Table 8: LoRA configuration used for GPT-2 adaptation.

Metric	Value
Accuracy	65.86%
Precision	65.45%
Recall	65.86%
F1-score	63.18%

Table 9: GPT-2 with LoRA performance on the test set.

in Section 5. The LoRA adapters were optimized while the original GPT-2 parameters remained frozen. Model performance on the validation set was monitored throughout training to reduce overfitting and select the best-performing checkpoint.

Because the majority of model parameters remained unchanged, the computational requirements were significantly lower than those associated with conventional full-model fine-tuning.

8.5 Experimental Results

The quantitative performance of GPT-2 with LoRA is summarized in Table 9.

Compared with GPT-2 Prompting, LoRA produced a substantial improvement across all evaluation metrics while updating only a very small fraction of the model parameters.

8.6 Discussion

Parameter-efficient fine-tuning significantly improved GPT-2’s ability to distinguish between customer support departments. Unlike prompt-based inference, LoRA enables GPT-2 to learn task-specific decision boundaries while preserving the general linguistic knowledge acquired during pre-training.

The most notable improvement was observed in the model’s ability to classify Billing and Payments and Technical Support tickets, indicating that lightweight adaptation successfully captured domain-specific language patterns associated with these departments. However, some confusion remained between General Inquiry and Sales and Pre-Sales, suggesting that these categories contain semantically overlapping customer requests that are more difficult to separate.

An important characteristic of LoRA is its computational efficiency. By updating only 0.24% of the model parameters, the approach achieved a meaningful performance improvement without requiring the computational cost of full transformer fine-tuning. Nevertheless, GPT-2 remains fundamentally an autoregressive language model originally designed for text generation rather than discriminative classification. This limitation motivates the evaluation of DistilBERT, an encoder-based transformer specifically optimized for supervised sequence classification.

9 DistilBERT Sequence Classification

9.1 Motivation

While GPT-2 and GPT-2 with LoRA demonstrated the potential of autoregressive language models for customer support email routing, these models were originally designed for text generation rather than discriminative classification. Their left-to-right decoding strategy limits the contextual information available when predicting class labels, particularly for customer requests containing long and semantically complex descriptions.

Encoder-based transformer models provide a different learning paradigm. Instead of predicting the next token, encoder architectures learn contextual representations by simultaneously attending to both preceding and succeeding words within a sequence. This bidirectional understanding makes encoder models particularly effective for supervised text classification tasks.

To investigate the benefits of supervised encoder-based learning, this study evaluates DistilBERT, a compact transformer model derived from BERT through knowledge distillation. DistilBERT retains much of BERT’s language understanding capability while substantially reducing model size and computational cost,

making it well suited for practical NLP applications.

9.2 DistilBERT Architecture

DistilBERT is an encoder-only transformer architecture developed through the knowledge distillation of the original BERT model. During pre-training, the smaller student model learns to reproduce the behavior of the larger teacher model while preserving most of its language understanding capability.

Compared with BERT, DistilBERT offers several practical advantages, including fewer parameters, faster inference, reduced memory consumption, competitive performance on NLP benchmarks, and strong contextual language understanding. Because DistilBERT processes the complete input sequence bidirectionally, it captures semantic relationships that are often difficult for autoregressive decoder models to represent effectively during classification.

9.3 Fine-Tuning Strategy

Unlike GPT-2 Prompting, which performs inference without training, and GPT-2 with LoRA, which updates only a small subset of parameters, DistilBERT was fully fine-tuned for sequence classification using the preprocessed customer support dataset.

The combined subject and body text constructed during preprocessing was tokenized using the DistilBERT tokenizer. The encoder representations were then optimized through supervised learning using the department labels described in Section 5. All transformer parameters were updated during training, enabling the model to learn task-specific semantic representations directly from customer support requests.

9.4 Training Configuration

Training was performed using the Hugging Face Trainer API. The model was trained using the same dataset splits employed for the previous experiments, ensuring a fair comparison across all three approaches.

9.5 Experimental Results

Table 11 summarizes the performance obtained by DistilBERT. Among the three evaluated approaches, DistilBERT achieved the highest performance across all evaluation metrics.

Parameter	Value
Base Model	DistilBERT
Task	Sequence Classification
Number of Classes	5
Optimizer	AdamW
Evaluation Strategy	Per Epoch
Dataset Split	Stratified
Framework	Hugging Face Trainer

Table 10: DistilBERT training configuration.

Metric	Value
Accuracy	72.17%
Precision	72.20%
Recall	72.17%
F1-score	70.07%

Table 11: DistilBERT performance on the test set.

9.6 Discussion

The experimental results demonstrate that DistilBERT provides the strongest overall performance for customer support email routing. The encoder-based architecture effectively captures contextual information from both the subject and body of each customer request, enabling more accurate discrimination between semantically related departments.

Compared with GPT-2 Prompting, DistilBERT achieved an improvement of approximately 15.7 percentage points in classification accuracy. Relative to GPT-2 with LoRA, DistilBERT further improved accuracy by approximately 6.3 percentage points, indicating that full supervised fine-tuning remains advantageous for this classification task.

The confusion matrix analysis showed a reduction in misclassification across most departments, particularly for Billing and Payments, Customer Service, and Technical Support. Some confusion persisted between General Inquiry and Sales and Pre-Sales, reflecting the semantic similarity of customer requests within these categories and the remaining class imbalance in the dataset.

Although DistilBERT updates all model parameters during training, it achieved the fastest inference among the evaluated models while delivering the highest predictive performance. This combination of efficiency and accuracy makes DistilBERT the most prac-

tical solution for intelligent customer support email routing within the scope of this study.

9.7 Key Findings

The DistilBERT experiments lead to several important observations. Full supervised fine-tuning provides the highest predictive performance. Bidirectional contextual representations substantially improve department classification. Encoder-based transformers outperform autoregressive decoder models on structured text classification tasks. DistilBERT offers an effective balance between computational efficiency and classification accuracy. Remaining prediction errors are primarily associated with semantically similar departments rather than limitations of the transformer architecture itself.

10 Experimental Results and Comparative Analysis

The objective of this section is to compare the three transformer-based approaches under identical experimental conditions and to analyze the trade-offs between predictive performance, computational efficiency, and training complexity.

10.1 Quantitative Performance Comparison

Table 12 summarizes the overall performance of the three evaluated models. DistilBERT consistently achieved the highest performance across all evaluation metrics, followed by GPT-2 with LoRA, while GPT-2 Prompting served as the baseline.

Model	Accuracy	Precision	Recall	F1-score
GPT-2 Prompting	56.50%	41.56%	56.50%	45.65%
GPT-2 + LoRA	65.86%	65.45%	65.86%	63.18%
DistilBERT	72.17%	72.20%	72.17%	70.07%

Table 12: Performance comparison of the three transformer-based approaches.

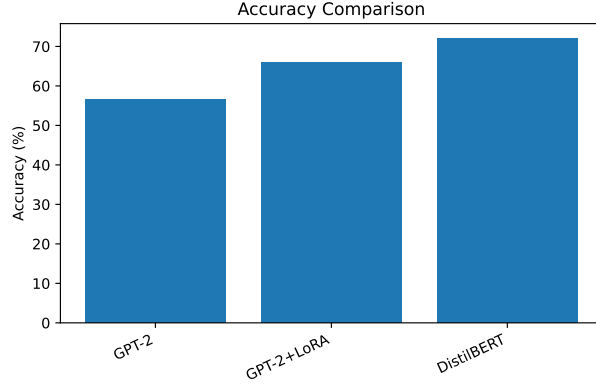


Figure 7: Accuracy comparison across the evaluated transformer models.

10.2 Performance Improvement Analysis

Compared with GPT-2 Prompting, GPT-2 with LoRA improved accuracy by 9.36 percentage points, while DistilBERT improved accuracy by 15.67 percentage points. Compared with GPT-2 with LoRA, DistilBERT achieved an additional 6.31 percentage point improvement. These results demonstrate that each successive adaptation strategy contributes measurable improvements in predictive performance.

10.3 Confusion Matrix Analysis

Although overall accuracy provides a useful summary, confusion matrices reveal how each model behaves at the class level.

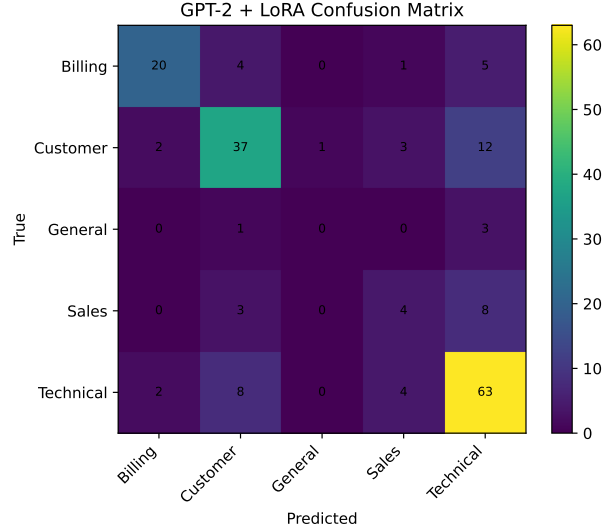


Figure 9: Confusion matrix for GPT-2 with LoRA.

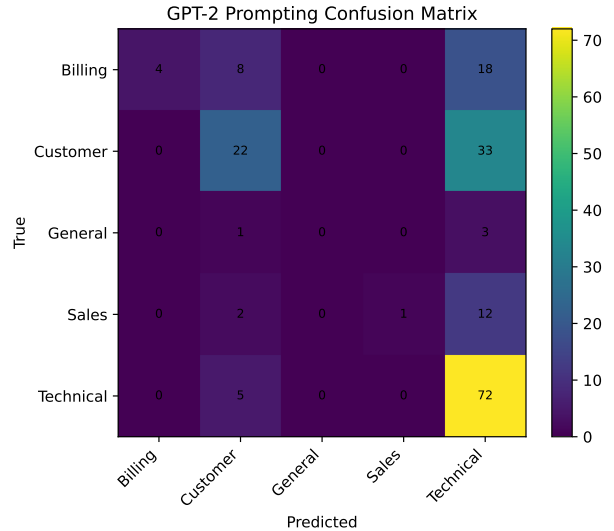


Figure 8: Confusion matrix for GPT-2 Prompting.

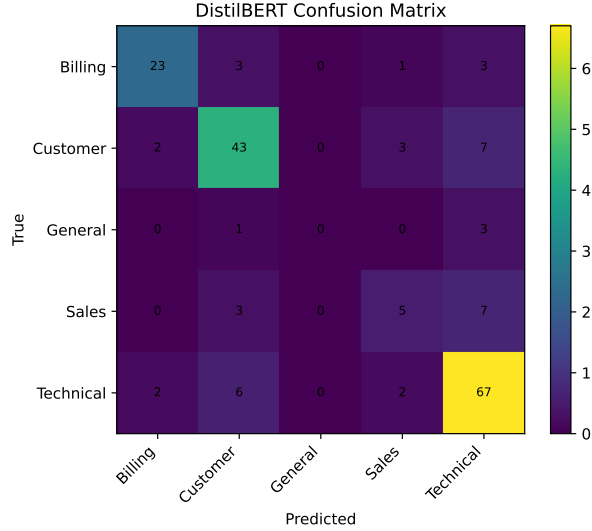


Figure 10: Confusion matrix for DistilBERT.

The GPT-2 baseline exhibited a strong tendency to predict Technical Support, reflecting both prompt ambiguity and class imbalance. Introducing LoRA reduced incorrect predictions for several classes, demonstrating that lightweight fine-tuning captured task-specific language patterns. DistilBERT produced the most balanced confusion matrix, with remaining errors concentrated among semantically related categories.

10.4 Computational Efficiency

Predictive performance alone does not fully characterize the suitability of a transformer model for real-world deployment. Computational cost and resource requirements are also important considerations.

GPT-2 Prompting requires no training but has the slowest inference because each prediction involves autoregressive text generation. GPT-2 with LoRA significantly reduces training cost by updating only 0.24% of the parameters while improving predictive performance. DistilBERT requires full supervised fine-tuning but achieves both the highest classification accuracy and the fastest inference among the trained models.

10.5 Comparative Discussion

GPT-2 Prompting offers the simplest deployment strategy because it requires no task-specific training. GPT-2 with LoRA provides an effective compromise between computational efficiency and predictive performance. DistilBERT delivers the strongest overall performance due to its bidirectional encoder architecture and supervised optimization objective.

The choice of model therefore depends on available computational resources, the quantity of labeled training data, and the desired balance between efficiency and predictive performance.

11 Discussion

The experimental evaluation demonstrates that transformer architecture and adaptation strategy have a substantial impact on the effectiveness of automatic customer support email routing. Although all three evaluated approaches were based on transformer models, their performance differed considerably due to differences in learning objectives, contextual representation, and training methodology.

The progression from GPT-2 Prompting to GPT-2 with LoRA and finally to DistilBERT illustrates how increasing levels of task-specific adaptation consistently improve predictive performance. The results therefore support the hypothesis that supervised transformer learning provides more effective semantic representations for structured text classification than prompt-based inference alone.

11.1 Interpretation of Experimental Findings

GPT-2 Prompting established an effective zero-shot baseline that required no additional training. This approach demonstrates that pre-trained language models possess general language understanding capabilities that can be applied directly to downstream tasks through carefully designed prompts. However, the results also reveal the limitations of relying solely on autoregressive text generation for multi-class classification. The generated responses occasionally deviated from the expected department labels, requiring rule-based post-processing and leading to lower classification accuracy.

Model	Training Required	Trainable Parameters	Inference Time
GPT-2 Prompting	No	0%	5.79 s / 50 samples
GPT-2 + LoRA	Yes	0.24%	0.84 s / 100 samples
DistilBERT	Yes	100%	0.22 s / 100 samples

Table 13: Computational comparison of the evaluated models.

Introducing Low-Rank Adaptation significantly improved GPT-2’s predictive performance while updating only 0.24% of the model parameters. This finding confirms that lightweight task-specific adaptation enables generative language models to learn domain-specific decision boundaries without incurring the computational cost of full fine-tuning. The improvement achieved through LoRA demonstrates the practical value of parameter-efficient learning, particularly in environments with limited computational resources.

DistilBERT achieved the strongest overall performance, suggesting that encoder-based transformer architectures remain particularly well suited for supervised sequence classification. Unlike autoregressive models that process text sequentially, DistilBERT simultaneously considers contextual information from both preceding and succeeding words. This bidirectional representation enables more accurate interpretation of customer requests and improves discrimination between support departments with similar semantic characteristics.

11.2 Impact of Dataset Characteristics

The characteristics of the customer support dataset also influenced model performance. Although restricting the dataset to five departments reduced the severity of class imbalance, noticeable differences in class frequencies remained. The confusion matrices indicate that the majority of remaining prediction errors occurred between General Inquiry and Sales and Pre-Sales, two categories whose textual descriptions frequently overlap.

Similarly, replacing missing textual fields with empty strings allowed the study to preserve a substantial number of customer support tickets that would otherwise have been discarded. Because transformer models effectively integrate contextual information across the entire input sequence, retaining partially

complete tickets provided additional training examples without introducing artificial textual content.

These observations emphasize that data preprocessing plays an essential role in transformer-based text classification and contributes significantly to the overall effectiveness of the learning pipeline.

11.3 Practical Implications

The findings of this study have practical implications for organizations seeking to automate customer support operations. For organizations with limited computational resources or limited access to labeled training data, prompt-based inference provides a simple and rapidly deployable solution. Although its predictive performance is lower than supervised approaches, it can serve as a useful baseline for rapid prototyping or low-volume applications.

When moderate computational resources are available, parameter-efficient fine-tuning through LoRA offers an attractive compromise between training efficiency and predictive performance. Updating only a small fraction of model parameters substantially reduces computational cost while providing meaningful improvements in classification accuracy.

For organizations prioritizing routing accuracy and operational reliability, fully supervised encoder-based models such as DistilBERT remain the preferred solution. Despite requiring full fine-tuning during training, DistilBERT achieved both the highest classification accuracy and the fastest inference among the evaluated models, making it suitable for production-scale customer support systems.

11.4 Research Implications

Beyond customer support automation, this comparative study contributes to the broader understanding of transformer adaptation strategies for practical Natural Language Processing applications. The results reinforce the growing consensus that parameter-efficient

fine-tuning represents a valuable intermediate solution between zero-shot prompting and full supervised learning.

Furthermore, the experiments demonstrate that selecting an appropriate transformer architecture is as important as selecting an adaptation strategy. While generative models such as GPT-2 excel at open-ended language generation, encoder-based architectures continue to provide superior performance for discriminative text classification tasks.

The methodology developed in this work can therefore be adapted to other document classification problems, including helpdesk ticket routing, technical support categorization, intent detection, customer feedback analysis, and automated document triage.

11.5 Lessons Learned

Several important lessons emerged during the implementation and evaluation of the proposed system. Careful dataset preprocessing significantly influences transformer performance. Prompt engineering alone is insufficient for high-quality multi-class classification. Parameter-efficient fine-tuning provides substantial performance improvements with minimal computational overhead. Full supervised learning remains the most effective strategy for structured text classification. Finally, fair comparison requires identical preprocessing, dataset splits, and evaluation procedures across all models.

Overall, the experimental evidence demonstrates that increasing levels of model adaptation consistently improve classification performance. GPT-2 Prompting establishes a meaningful zero-shot baseline, GPT-2 with LoRA illustrates the effectiveness of parameter-efficient fine-tuning, and DistilBERT confirms the advantages of supervised encoder-based learning for customer support email routing.

12 Limitations

Despite the encouraging performance achieved by the evaluated transformer models, several limitations should be acknowledged when interpreting the findings of this study.

12.1 Dataset Scope

The original Customer Support Tickets dataset contains 52 support departments. This study intentionally focused on five representative departments to construct a balanced and computationally manageable multi-class classification problem. While this enabled a fair comparison of the evaluated transformer models, it does not fully capture the complexity of large-scale industrial customer support systems.

12.2 Class Imbalance

Although department selection reduced the degree of imbalance present in the original dataset, noticeable differences in class frequencies remained. Most prediction errors occurred between General Inquiry and Sales and Pre-Sales. Future work could investigate weighted loss functions, focal loss, oversampling, or synthetic data generation.

12.3 Model Selection

This study evaluated GPT-2 Prompting, GPT-2 with LoRA, and DistilBERT. More recent architectures such as RoBERTa, DeBERTa, and domain-adapted transformer models may achieve stronger classification performance.

12.4 Hyperparameter Optimization

The experiments employed a consistent training configuration to ensure a fair comparison. Extensive hyperparameter optimization was beyond the scope of this work and may further improve predictive performance.

12.5 Generalizability

The evaluated models were trained and tested using a single publicly available dataset. Additional evaluation on multiple datasets and industrial customer support systems would be necessary to fully assess the generalizability of the proposed approaches.

12.6 Summary

These limitations define the boundaries within which the reported results should be interpreted while identifying meaningful directions for future investigation.

13 Future Work

Several opportunities remain for extending and improving the proposed system.

13.1 Larger Transformer Models

Future work could evaluate RoBERTa, DeBERTa, ModernBERT, and other recent transformer architectures to determine whether they further improve customer support ticket classification.

13.2 Advanced PEFT Methods

Alternative parameter-efficient fine-tuning approaches such as QLoRA, AdaLoRA, IA³, and prompt tuning could be compared under the same evaluation framework.

13.3 Class Imbalance

Future studies may investigate weighted losses, focal loss, oversampling, and data augmentation to improve minority-class performance.

13.4 Multilingual Customer Support

The original dataset contains both English and German tickets. Future work could extend the pipeline using multilingual transformer models such as mBERT or XLM-RoBERTa.

13.5 Explainable AI

Explainability techniques including SHAP, LIME, and attention visualization could improve transparency and user trust.

13.6 Real-Time Deployment

The best-performing model could be deployed through a REST API for automatic routing of incoming customer support requests in production environments.

13.7 Summary

The methodology developed in this study provides a strong foundation for future investigations into efficient, accurate, and scalable transformer-based customer support automation.

14 Conclusion

This paper presented a comprehensive comparative evaluation of three transformer-based approaches for intelligent customer support email routing: GPT-2 Prompting, GPT-2 with Low-Rank Adaptation (LoRA), and DistilBERT.

A complete NLP pipeline was developed using the Customer Support Tickets dataset, including exploratory data analysis, preprocessing, transformer-based model development, quantitative evaluation, and comparative analysis under identical experimental conditions.

The experiments demonstrated a clear progression in predictive performance. GPT-2 Prompting achieved 56.50% accuracy without supervised training. GPT-2 with LoRA improved performance to 65.86% while updating only 0.24% of the model parameters. DistilBERT achieved the strongest overall performance with an accuracy of 72.17%.

The results show that prompt-based inference provides a useful baseline, parameter-efficient fine-tuning offers an attractive compromise between efficiency and performance, and supervised encoder-based transformers remain the most effective solution for customer support email classification when labeled training data are available.

Overall, this work contributes a reproducible experimental framework for evaluating transformer adaptation strategies and provides practical guidance for deploying transformer models in intelligent customer support systems.

References

- [1] Jacob Devlin, Ming-Wei Chang, Kenton Lee, and Kristina Toutanova. Bert: Pre-training of deep bidirectional transformers for language understanding. In *NAACL*, 2019.
- [2] Edward J. Hu et al. Lora: Low-rank adaptation of large language models. In *ICLR*, 2022.
- [3] Victor Sanh et al. Distilbert. In *NeurIPS Workshop*, 2019.
- [4] Ashish Vaswani et al. Attention is all you need. In *NeurIPS*, 2017.